

- ▶ **Apollo 15:** David Scott and Jim Irwin drove a total of 27.8 km in 3 hours, 2 minutes. The longest single traverse was 12.5 km and the maximum range from the LM was 5.0 km.
- ▶ **Apollo 16:** John Young and Charles Duke drove 26.7 km in 3 hours 26 minutes of driving. The longest traverse was 11.6 km and the LRV reached a distance of 4.5 km from the LM.
- ▶ **Apollo 17:** Gene Cernan and Harrison Schmitt drove 35.9 km in 4 hours 26 minutes. The longest traverse was 20.1 km and the greatest range from the LM was 7.6 km.

The Lunar Roving Vehicle had a mass of 210 kg and was designed to hold a payload of an additional 490 kg on the lunar surface. The frame was 3.1 metres long with a wheelbase of 2.3 metres. The maximum height was 1.14 metres. The frame was made of aluminium alloy and consisted of a 3-part chassis which was hinged in the centre so it could be folded up and hung in the Lunar Module quad 1 bay. It had two side-by-side foldable seats made of tubular aluminium with nylon webbing and aluminium floor panels. An armrest was mounted between the seats, and each seat had adjustable footrests and a Velcro seatbelt. A large mesh dish antenna was mounted on a mast on the front centre of the rover. The suspension consisted of a double horizontal wishbone with upper and lower torsion bars and a damper unit between the chassis and upper wishbone. Fully loaded, the LRV had a ground clearance of 14 inches.

The wheels consisted of a spun aluminium hub. The tires were made out of zinc-coated woven steel, measuring 32 inches in diameter and 9 inches wide. Inside the tire was a 25-inch-diameter bump stop frame to protect the hub. Dust guards were mounted above the wheels. Each wheel had its own electric drive and a mechanical brake unit. Manoeuvring capability was provided through the use of front and rear steering motors. Each series wound DC steering motor was capable of 0.1 hp. Both sets of wheels would turn in opposite directions, giving a steering radius of 3.1 metres, or could be decoupled so only one set would be used for steering.

Power was provided by two 36-volt silver-zinc potassium hydroxide non-rechargeable batteries. These were used to power the drive and steering motors and also a 36 volt utility outlet mounted on front of the LRV to power the communications relay unit or the TV camera. Passive thermal controls kept the batteries within an optimal temperature range.

A T-shaped hand controller situated between the two seats controlled the four drive motors, two steering motors and brakes. Moving the stick forward powered the LRV forward; left and right turned the vehicle left or